RBD_Mixed_Model_Analysis

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Linear Mixed Model for a Randomized Block Design

A **linear mixed model** separates what we want to compare from what we only want to account for:

- **Fixed effect:** the exact levels we care about and want to compare. Here, the **treatments**.
- **Random effect:** levels that are a *sample* from a larger population, which we do not care about individually. Here, the **blocks**.

The question changes from "how much higher is Block 4 than Block 3?" to "how much do blocks vary in general, and can that variation be removed so treatments are compared fairly?"

---

1. When Do We Need a Mixed Model?

Situation	Method
Complete, balanced, normal data	Normal ANOVA (mixed model gives the same answer)
Missing plots or unbalanced data	**Mixed model**
Several random factors (blocks across locations or seasons)	**Mixed model**
Repeated measurements on the same plot over time	**Mixed model**
Clearly non-normal data, complete blocks	Friedman test

Why missing values are the main reason

Suppose a plot in Block 4 (a high-yielding block) is lost.

- **Raw treatment mean:** that treatment's average falls only because its plot was missing from a good block. The comparison is unfair.
- **Mixed model:** it knows Block 4 is usually high, so it adjusts. The estimated treatment mean is what the treatment would have yielded across all blocks.

In this file, the dataset has missing values, so the mixed model is the appropriate analysis.

---

2. Data

Load the data

```
datamm <- read.csv("rbd_mixed_model_missing_data.csv")

head(datamm)
```

```
    block   treatment yield_t_ha
1 Block_1     Control       3.19
2 Block_1         NPK       3.82
3 Block_1     Organic       3.69
4 Block_1 NPK_Organic       4.35
5 Block_1        Urea       4.04
6 Block_1     Compost       3.97
```

Convert block and treatment to factors

```
datamm$block     <- as.factor(datamm$block)
datamm$treatment <- as.factor(datamm$treatment)
```

Check the missing values

```
colSums(is.na(datamm))
```

```
     block  treatment yield_t_ha
         0          0          4
```

```
# Which rows are missing?
datamm[is.na(datamm$yield_t_ha), ]
```

```
     block   treatment yield_t_ha
9  Block_2     Organic         NA
14 Block_3         NPK         NA
23 Block_4        Urea         NA
28 Block_5 NPK_Organic         NA
```

Keep only the observed plots

The model ignores rows with a missing response. We remove them explicitly, so it is clear which data are used:

```
datamm_obs <- na.omit(datamm)

nrow(datamm)
```

```
[1] 30
```

```
nrow(datamm_obs)
```

```
[1] 26
```

---

3. Why Raw Means Are Not Reliable Here

Raw treatment means ignore which blocks the plots came from:

```
tapply(datamm_obs$yield_t_ha, datamm_obs$treatment, mean)
```

```
    Compost     Control         NPK NPK_Organic     Organic        Urea
     4.0620      3.3840      4.0200      4.5425      3.8200      4.0350
```

We will compare these with the model-adjusted means in Section 7.

---

4. Model Building

The mixed model is:

$$Y_{ij} = \mu + \tau_i + b_j + \epsilon_{ij}, \qquad b_j \sim N(0, \sigma_b^2), \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

In R, `(1 | block)` means "each block gets its own random intercept".

```r
library(lme4)
```

```
Loading required package: Matrix
```

```r
library(lmerTest)
```

```
Warning: package 'lmerTest' was built under R version 4.4.3
```

```
Attaching package: 'lmerTest'
```

```
The following object is masked from 'package:lme4':

    lmer
```

```
The following object is masked from 'package:stats':

    step
```

```r
model_mm <- lmer(
  yield_t_ha ~ treatment + (1 | block),
  data = datamm_obs
)

summary(model_mm)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: yield_t_ha ~ treatment + (1 | block)
   Data: datamm_obs

REML criterion at convergence: -17.1

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.89176 -0.55305  0.02547  0.47098  1.54397

Random effects:
 Groups   Name        Variance Std.Dev.
```

```
 block    (Intercept) 0.0357   0.18894
 Residual             0.0087   0.09327
Number of obs: 26, groups:  block, 5

Fixed effects:
                     Estimate Std. Error       df t value Pr(>|t|)
(Intercept)           4.06200    0.09423  5.56669  43.107 3.04e-08 ***
treatmentControl     -0.67800    0.05899 15.99413 -11.493 3.85e-09 ***
treatmentNPK         -0.07139    0.06325 16.02914  -1.129  0.27568
treatmentNPK_Organic  0.46025    0.06325 16.02914   7.276 1.84e-06 ***
treatmentOrganic     -0.23684    0.06325 16.02914  -3.744  0.00176 **
treatmentUrea         0.05100    0.06325 16.02914   0.806  0.43191
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) trtmnC trtNPK tNPK_O trtmnO
trtmntCntrl -0.313
treatmntNPK -0.292  0.466
trtmntNPK_O -0.292  0.466  0.429
trtmntOrgnc -0.292  0.466  0.429  0.429
treatmentUr -0.292  0.466  0.429  0.429  0.429
```

How to read the output:

- **Random effects:** the variance of `block` and the residual variance.
- **Fixed effects:** treatment coefficients relative to the reference treatment (the first level alphabetically).

Check for a singular fit

```
isSingular(model_mm)
```

```
[1] FALSE
```

- **FALSE:** the model is fine.
- **TRUE:** the block variance is estimated as essentially zero, meaning blocks explained almost nothing. The treatment results are still usable, blocking was not effective.

---

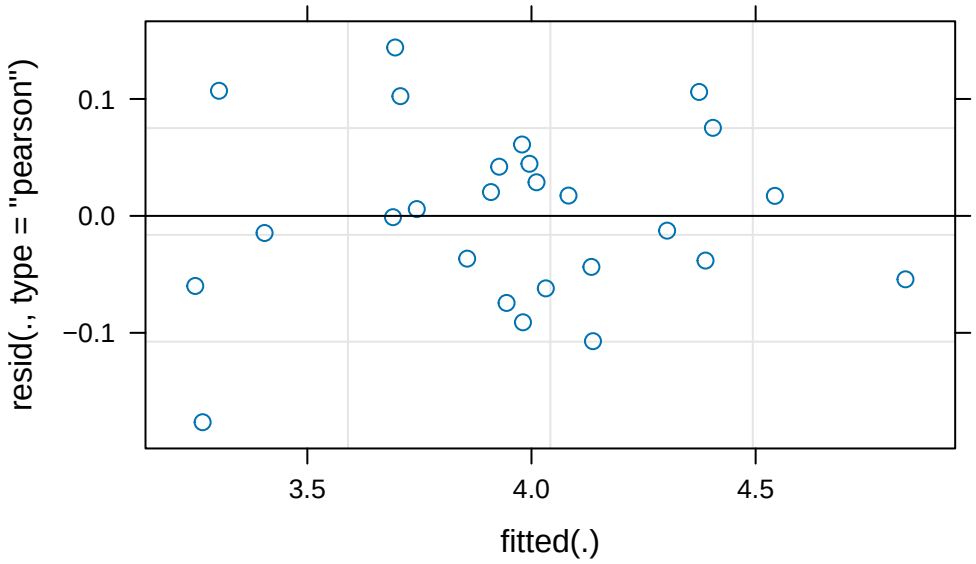
5. Model Diagnostics

The assumptions are:

1. Residuals are approximately normal.
2. Residual variance is constant.
3. Block random effects are approximately normal.
4. Observations are independent after accounting for block.

5.1 Residuals vs fitted

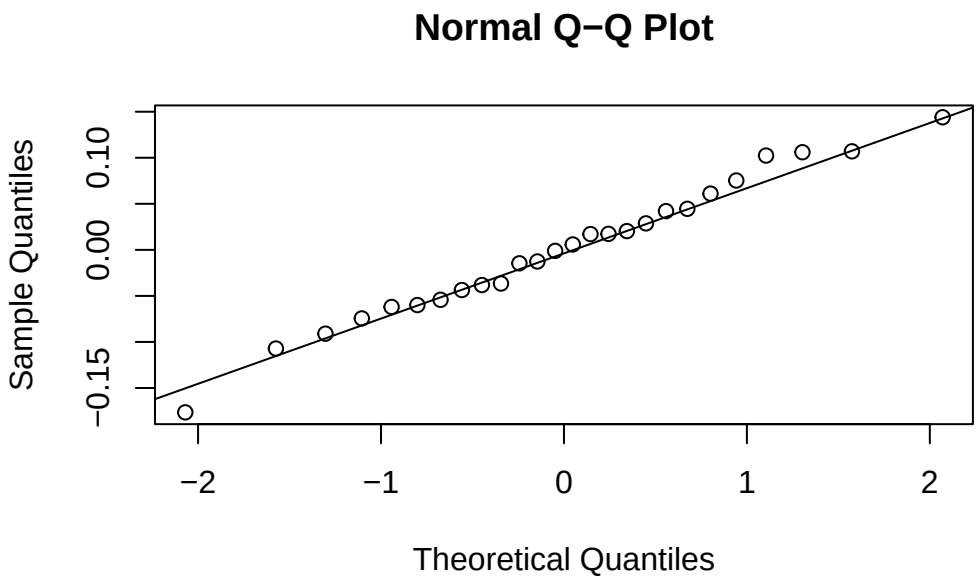
```
plot(model_mm)
```



Look for a random spread with no funnel or curve.

5.2 Normality of residuals

```
qqnorm(resid(model_mm))
qqline(resid(model_mm))
```




```
shapiro.test(resid(model_mm))
```

```
    Shapiro-Wilk normality test

data:  resid(model_mm)
W = 0.98759, p-value = 0.9831
```

- **p > 0.05:** no evidence against normality.
- **p < 0.05:** check the Q-Q plot. If the deviation is clear, try a log transformation (`log(yield_t_ha)`) and refit.

5.3 Homogeneity of variance

```
library(car)
```

```
Warning: package 'car' was built under R version 4.4.3
```

```
Loading required package: carData
```

```r
leveneTest(resid(model_mm) ~ datamm_obs$treatment)
```

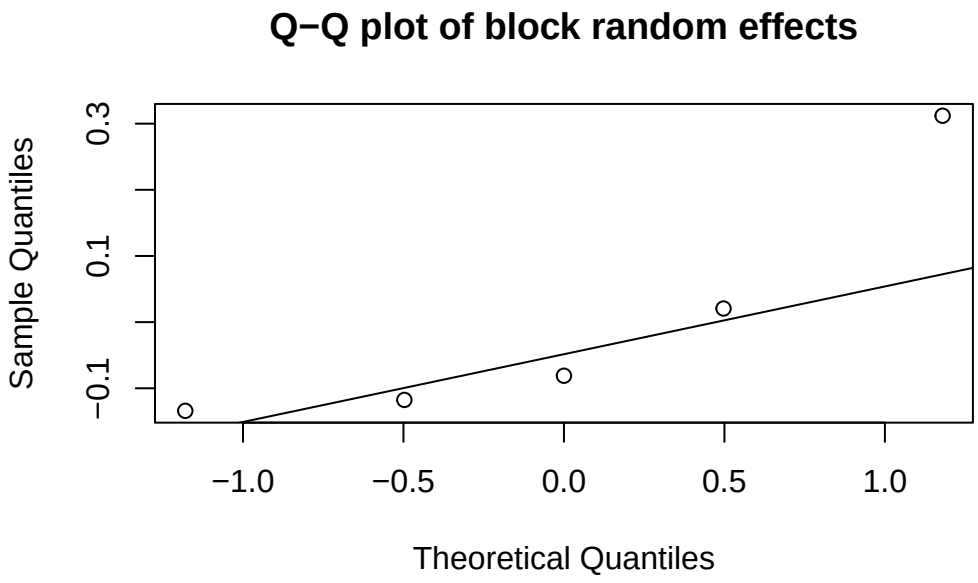
```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  5  1.2202  0.336
      20
```

With so few observations per treatment, this test has low power, so the residuals vs fitted plot matters more.

5.4 Normality of the block random effects

```r
re_block <- ranef(model_mm)$block[, 1]

qqnorm(re_block, main = "Q-Q plot of block random effects")
qqline(re_block)
```



With only a few blocks, this is only a rough check.

5.5 Diagnostic decision

Decision: There are no serious violations, proceed with the mixed model results.

---

6. Mixed Model ANOVA

```
anova(model_mm)
```

```
Type III Analysis of Variance Table with Satterthwaite's method
          Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
treatment 3.0917 0.61834     5 16.042  71.076 2.268e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- **Null hypothesis (H):** all treatment means are equal.
- **Alternative hypothesis (H):** at least one treatment mean differs.
- **Significance level:** $= 0.05$

Decision: reject H

How much variation is due to blocks?

```
vc <- as.data.frame(VarCorr(model_mm))
vc
```

```
       grp        var1 var2        vcov      sdcor
1    block (Intercept) <NA> 0.035698088 0.18893938
2 Residual        <NA> <NA> 0.008699782 0.09327262
```

```
var_block <- vc$vcov[vc$grp == "block"]
var_resid <- vc$vcov[vc$grp == "Residual"]

# Intraclass correlation (ICC)
var_block / (var_block + var_resid)
```

```
[1] 0.8040496
```

The proportion is the **intraclass correlation (ICC)**. A large value means plots in the same block are similar, so blocking was effective. A value near 0 means blocking removed little variation.

---

7. Adjusted Means and Post-hoc Analysis

`emmeans` gives **estimated marginal means**, which are the model-adjusted treatment means (corrected for the missing plots).

```
library(emmeans)
```

```
Warning: package 'emmeans' was built under R version 4.4.3
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```
emm_mm <- emmeans(model_mm, pairwise ~ treatment, adjust = "tukey")
emm_mm
```

```
$emmeans
 treatment   emmean     SE   df lower.CL upper.CL
 Compost       4.06 0.0942 5.57     3.83     4.30
 Control       3.38 0.0942 5.57     3.15     3.62
 NPK           3.99 0.0970 6.19     3.76     4.23
 NPK_Organic   4.52 0.0970 6.19     4.29     4.76
 Organic       3.83 0.0970 6.19     3.59     4.06
 Urea          4.11 0.0970 6.19     3.88     4.35

Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
 contrast              estimate     SE   df t.ratio p.value
 Compost - Control       0.6780 0.0590 16.0  11.493  <.0001
 Compost - NPK           0.0714 0.0633 16.0   1.128  0.8628
 Compost - NPK_Organic  -0.4602 0.0633 16.0  -7.271  <.0001
 Compost - Organic       0.2368 0.0633 16.0   3.742  0.0182
```

```
 Compost - Urea         -0.0510 0.0633 16.0  -0.806  0.9623
 Control - NPK          -0.6066 0.0633 16.0  -9.584  <.0001
 Control - NPK_Organic  -1.1382 0.0633 16.0 -17.983  <.0001
 Control - Organic      -0.4412 0.0633 16.0  -6.970  <.0001
 Control - Urea         -0.7290 0.0633 16.0 -11.517  <.0001
 NPK - NPK_Organic      -0.5316 0.0677 16.1  -7.854  <.0001
 NPK - Organic           0.1655 0.0677 16.1   2.444  0.1986
 NPK - Urea             -0.1224 0.0677 16.1  -1.808  0.4878
 NPK_Organic - Organic   0.6971 0.0677 16.1  10.299  <.0001
 NPK_Organic - Urea      0.4093 0.0677 16.1   6.046  0.0002
 Organic - Urea         -0.2878 0.0677 16.1  -4.253  0.0066

Degrees-of-freedom method: kenward-roger
P value adjustment: tukey method for comparing a family of 6 estimates
```

Compact letter display

```
library(multcomp)
```

```
Warning: package 'multcomp' was built under R version 4.4.3
```

```
Loading required package: mvtnorm
```

```
Loading required package: survival
```

```
Loading required package: TH.data
```

```
Loading required package: MASS
```

```

Attaching package: 'TH.data'
```

```
The following object is masked from 'package:MASS':

    geyser
```

```r
library(multcompView)
```

```
Warning: package 'multcompView' was built under R version 4.4.3
```

```r
multcomp::cld(emm_mm$emmeans, adjust = "tukey", Letters = letters)
```

```
Note: adjust = "tukey" was changed to "sidak"
because "tukey" is only appropriate for one set of pairwise comparisons

 treatment   emmean     SE   df lower.CL upper.CL .group
 Control       3.38 0.0942 5.57     3.01     3.76  a
 Organic       3.83 0.0970 6.19     3.46     4.19   b
 NPK           3.99 0.0970 6.19     3.62     4.36   bc
 Compost       4.06 0.0942 5.57     3.69     4.44    c
 Urea          4.11 0.0970 6.19     3.75     4.48    c
 NPK_Organic   4.52 0.0970 6.19     4.15     4.89     d

Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95
Conf-level adjustment: sidak method for 6 estimates
P value adjustment: tukey method for comparing a family of 6 estimates
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```

Treatments sharing the same letter are **not statistically different**.

Raw means vs adjusted means

```r
raw_means <- tapply(datamm_obs$yield_t_ha, datamm_obs$treatment, mean)
adj_means <- summary(emm_mm$emmeans)

data.frame(
  treatment = adj_means$treatment,
  raw_mean = as.numeric(raw_means[as.character(adj_means$treatment)]),
  adjusted_mean = adj_means$emmean
)
```

```
    treatment raw_mean adjusted_mean
1     Compost   4.0620      4.062000
2     Control   3.3840      3.384000
3         NPK   4.0200      3.990612
4 NPK_Organic   4.5425      4.522249
5     Organic   3.8200      3.825156
6        Urea   4.0350      4.112997
```

Treatments that lost a plot from a high or low block will show the biggest difference between the two columns. The adjusted mean is the fairer one to report.

---

8. Post-hoc Interpretation

(Write after running: give the ranking by adjusted mean and state which treatments differ at = 0.05.)

```
Treatment_A > Treatment_B > ... > Treatment_F
```

---

9. Agricultural Interpretation

(Write after running: highest and lowest treatments, their adjusted means in t/ha, and the size of the difference.)

Cost, fertilizer rate, economic return, and environmental effects should also be considered before making a recommendation.

Report that the **mixed model was used because of missing plots**, so treatment means are adjusted for block differences.